Mid-term examination on Logic Circuit Design

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1 Binary numbers

Question. If $N = b_{n-1}b_{n-2}\ldots b_0$ is an n-bit binary number, what is $2N$ and $\lfloor N/2 \rfloor$?

2 Boolean algebra

Question. Using truth tables, prove the distributive laws

$$A + BC \overset{1}{=} (A+B)(A+C)$$
$$A(B+C) \overset{2}{=} AB + AC$$

Question. Given the distributive laws and the following

$$1 + A \overset{3}{=} 1 \qquad 0 \cdot A \overset{4}{=} 0 \qquad 0 + A \overset{5}{=} A \qquad 1 \cdot A \overset{6}{=} A$$
$$A + A \overset{7}{=} A \qquad A \cdot A \overset{8}{=} A \qquad A + \overline{A} \overset{9}{=} 1 \qquad A \cdot \overline{A} \overset{10}{=} 0 \qquad \overline{\overline{A}} \overset{11}{=} A$$

prove the absorption laws

$$A + AB = A \tag{1}$$
$$A(A+B) = A \tag{2}$$

without truth tables.

3 Addition of BCD numbers

Question. State any overflow or underflow.

(a)
$$\begin{array}{r} 0111 \;\; 0101 \;\; 1001 \;\; 0000 \\ +\; 0010 \;\; 1001 \;\; 0011 \;\; 0101 \\ \hline \end{array}$$

(b)
$$\begin{array}{r} 0011 \;\; 0110 \;\; 0001 \;\; 0111 \\ +\; 0001 \;\; 0100 \;\; 0110 \;\; 1000 \\ \hline \end{array}$$

(c)
$$\begin{array}{r} 1001 \;\; 0101 \;\; 1000 \;\; 0111 \\ +\; 0011 \;\; 0101 \;\; 0110 \;\; 0010 \\ \hline \end{array}$$